

Practice Problems on Linear Transformations

Linear Algebra - 24DS

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Spring 2026

What is a Linear Transformation?

Quick Recap

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if for all vectors $\mathbf{u}, \mathbf{v}$ and scalars c :

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$$

$$T(c\mathbf{u}) = c T(\mathbf{u})$$

Equivalently (combined property):

$$T(c_1\mathbf{u} + c_2\mathbf{v}) = c_1 T(\mathbf{u}) + c_2 T(\mathbf{v})$$

Two necessary conditions (quick checks):

- 1 $T(\mathbf{0}) = \mathbf{0}$ (zero vector maps to zero vector)
- 2 No constant terms or nonlinear operations (squares, products, etc.)

Quick Checks Before Diving In

Linear transformations must satisfy:

$$T(x_1, x_2, \dots) = (\text{linear combinations of } x_1, x_2, \dots)$$

Linear	Not Linear
$T(x, y) = (2x, y - x)$	$T(x, y) = (x^2, y)$
$T(x, y, z) = (x + y, z)$	$T(x, y) = (x + 1, y)$
$T(x, y) = (3x, 0)$	$T(x, y) = (\sin x, y)$
$T(x, y, z) = (x - y, y + z, z - x)$	$T(x, y) = (xy, x)$

Key: Each output component must be a **linear combination** of inputs (no constants, no powers, no products, no trig).

Example 1: Linear Transformation

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + y \\ x - 3y \end{pmatrix}$$

Check 1: $T(\mathbf{0}) = T(0, 0) = (0, 0)$

Check 2 (Additivity):

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(x_1 + x_2, y_1 + y_2) \\ &= \begin{pmatrix} 2(x_1 + x_2) + (y_1 + y_2) \\ (x_1 + x_2) - 3(y_1 + y_2) \end{pmatrix} \\ &= \begin{pmatrix} 2x_1 + y_1 \\ x_1 - 3y_1 \end{pmatrix} + \begin{pmatrix} 2x_2 + y_2 \\ x_2 - 3y_2 \end{pmatrix} \\ &= T(\mathbf{u}) + T(\mathbf{v}) \quad \checkmark \end{aligned}$$

Check 3 (Homogeneity): Similar verification

T is linear!

Example 2: Linear Transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - y + 2z \\ 3x + z \end{pmatrix}$$

Matrix representation:

$$T(\mathbf{x}) = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Any transformation of the form $T(\mathbf{x}) = A\mathbf{x}$ is linear!

Linear (matrix transformation)

Example 3: Linear Transformation

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ -y \\ x + 3y \end{pmatrix}$$

Matrix form:

$$T(\mathbf{x}) = \begin{bmatrix} 2 & 0 \\ 0 & -1 \\ 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Verification:

$$T(c\mathbf{x}) = \begin{pmatrix} 2cx \\ -cy \\ cx + 3cy \end{pmatrix} = c \begin{pmatrix} 2x \\ -y \\ x + 3y \end{pmatrix} = cT(\mathbf{x})$$

Additivity also holds.

Linear

Example 4: Linear Transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + y \\ y + z \\ z + x \end{pmatrix}$$

Matrix:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Test homogeneity:

$$T(kx, ky, kz) = (kx + ky, ky + kz, kz + kx) = k(x + y, y + z, z + x) = kT(x, y, z)$$

Linear

Example 5: Linear Transformation

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$$

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2x_1 - x_2 + x_4 \\ x_2 + 3x_3 \\ x_1 + x_3 - 2x_4 \end{pmatrix}$$

Matrix: 4×3 matrix with rows corresponding to outputs.

$$A = \begin{bmatrix} 2 & -1 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 1 & 0 & 1 & -2 \end{bmatrix}$$

Since $T(\mathbf{x}) = A\mathbf{x}$, it is **linear**.

Linear

Example 6: Zero Transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (0, 0, 0) \quad \text{for all } (x, y, z)$$

Check:

- $T(\mathbf{u} + \mathbf{v}) = \mathbf{0} = \mathbf{0} + \mathbf{0} = T(\mathbf{u}) + T(\mathbf{v})$
- $T(c\mathbf{u}) = \mathbf{0} = c\mathbf{0} = cT(\mathbf{u})$

Linear (trivial linear transformation) This is the zero matrix transformation.

Example 7: Identity Transformation

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$T(x_1, x_2, \dots, x_n) = (x_1, x_2, \dots, x_n)$$

Check:

- $T(\mathbf{u} + \mathbf{v}) = \mathbf{u} + \mathbf{v} = T(\mathbf{u}) + T(\mathbf{v})$
- $T(c\mathbf{u}) = c\mathbf{u} = cT(\mathbf{u})$

Linear (identity matrix)

Non-Example 1: Constant Term

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x + 1, y + 2)$$

Fails zero check:

$$T(0, 0) = (1, 2) \neq (0, 0) \Rightarrow \text{NOT linear!}$$

Reason: Any linear transformation must satisfy $T(\mathbf{0}) = \mathbf{0}$.

■ Because $T(\mathbf{0}) = T(0 \cdot \mathbf{v}) = 0 \cdot T(\mathbf{v}) = \mathbf{0}$

Non-Example 2: Square Term

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x^2, y)$$

Test homogeneity:

$$T(cx, cy) = ((cx)^2, cy) = (c^2x^2, cy)$$

$$cT(x, y) = (cx^2, cy)$$

For $c = 2$, $T(2x, 2y) = (4x^2, 2y)$ but $2T(x, y) = (2x^2, 2y)$

$$4x^2 \neq 2x^2 \quad \text{unless } x = 0$$

NOT linear (squaring is nonlinear)

Non-Example 3: Product of Variables

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x, y, z) = (xy, z)$$

Test additivity:

$$\begin{aligned} T((x_1, y_1, z_1) + (x_2, y_2, z_2)) &= ((x_1 + x_2)(y_1 + y_2), z_1 + z_2) \\ &= (x_1y_1 + x_1y_2 + x_2y_1 + x_2y_2, z_1 + z_2) \end{aligned}$$

$$T(x_1, y_1, z_1) + T(x_2, y_2, z_2) = (x_1y_1 + x_2y_2, z_1 + z_2)$$

These are not equal (extra cross terms $x_1y_2 + x_2y_1$).

NOT linear

Non-Example 4: Trigonometric Function

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (\sin x, y)$$

Test additivity:

$$T(x_1 + x_2, y_1 + y_2) = (\sin(x_1 + x_2), y_1 + y_2)$$

$$T(x_1, y_1) + T(x_2, y_2) = (\sin x_1 + \sin x_2, y_1 + y_2)$$

$\sin(x_1 + x_2) \neq \sin x_1 + \sin x_2$ in general (e.g., $x_1 = x_2 = \pi/2$: $\sin \pi = 0$, $\sin \pi/2 + \sin \pi/2 = 1 + 1 = 2$)

NOT linear

Non-Example 5: Absolute Value

$$T : \mathbb{R} \rightarrow \mathbb{R}$$

$$T(x) = |x|$$

Test homogeneity:

$$T(-1) = |-1| = 1, \quad -T(1) = -|1| = -1$$

$$T(-1) \neq -T(1)$$

NOT linear For $\mathbb{R}^2$: $T(x, y) = |x|$ also not linear.

Non-Example 6: Division by Variable

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$T(x, y) = \frac{x}{y} \quad (y \neq 0)$$

Test homogeneity:

$$T(cx, cy) = \frac{cx}{cy} = \frac{x}{y}$$

$$cT(x, y) = c \cdot \frac{x}{y}$$

These are equal only if $c = 1$ or $x = 0$.

Also fails domain issues (not defined for all vectors) and fails additivity.

NOT linear

Example 8: Projection onto x-axis

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x, 0, 0)$$

Matrix:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Check:

$$T(cx, cy, cz) = (cx, 0, 0) = c(x, 0, 0) = cT(x, y, z)$$

Additivity holds similarly.

Linear (projection)

Example 9: Reflection across $y = x$

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (y, x)$$

Matrix:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Check:

$$T(cx, cy) = (cy, cx) = c(y, x) = cT(x, y)$$

$$T(x_1+x_2, y_1+y_2) = (y_1+y_2, x_1+x_2) = (y_1, x_1) + (y_2, x_2) = T(x_1, y_1) + T(x_2, y_2)$$

Linear (reflection)

Example 10: Rotation by 90°

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (-y, x)$$

Matrix:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Check:

$$T(cx, cy) = (-cy, cx) = c(-y, x) = cT(x, y)$$

Additivity:

$$T(x_1 + x_2, y_1 + y_2) = (-(y_1 + y_2), x_1 + x_2) = (-y_1, x_1) + (-y_2, x_2)$$

Linear (rotation)

Summary: How to Spot Linearity

Linear Transformations	Non-Linear Transformations
$T(x) = ax$	$T(x) = ax + b$ (constant term)
$T(x, y) = (ax + by, cx + dy)$	$T(x) = x^2$ (power $\neq 1$)
$T(\mathbf{x}) = A\mathbf{x}$	$T(x, y) = xy$ (product)
Projections	$T(x) = \sin x, e^x, x $
Reflections	$T(x) = 1/x$ (division)
Rotations	$T(x, y) = (\sqrt{x}, y)$ (root)
Zero transformation	$T(x, y) = (\max(x, y), y)$

Check: $T(\mathbf{0}) = \mathbf{0}$ and no nonlinear operations!

Practice Problems

Determine if each is linear. Justify.

1 $T(x, y) = (2x - y, 0)$

2 $T(x, y, z) = (x + 2, y - z, z)$

3 $T(x, y) = (x, y^2)$

4 $T(x, y, z) = (z, y, x)$ (swap first and last)

5 $T(x, y) = (|x|, |y|)$

6 $T(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_3 + x_4, 0)$

7 $T(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$ (rotation by θ)

8 $T(x, y) = (\sqrt{x^2 + y^2}, 0)$

Remember: $T(\mathbf{0}) = \mathbf{0}$ is necessary but NOT sufficient!